SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
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COURSE OUTCOME COVERED

CO5:
Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

1. **A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.**

ANSWER:

Network Design for a Remote Rural Branch Office

Since the budget is limited, the solution must provide a balance between bandwidth, reliability, security, cost and scalability.

The major constraints are:

- High-speed wired Internet is unavailable.
- Cloud applications require reliable Internet connectivity.
- Video conferencing requires sufficient bandwidth and low latency.
- Communication with headquarters must be secure.

- The budget is limited.
- Business operations must continue even if the primary connection fails.
- The network should support future expansion.

Proposed Solution:

- A **4G/5G Fixed Wireless Access connection** can be used as the primary Internet connection because it does not require wired infrastructure and can provide suitable bandwidth in rural areas with cellular coverage.
- A **satellite Internet connection** can be used as a backup. A **dual-WAN router** can automatically switch to the backup connection if the primary connection fails.

Recommended Devices

- 4G/5G Router – Primary Internet connectivity.
- Satellite Modem – Backup Internet connection.
- Dual-WAN Router – Manages both connections and failover.
- Firewall – Protects the network from unauthorized traffic.
- Managed Switch – Connects office devices.
- Wireless Access Points – Provides employee Wi-Fi.
- UPS – Keeps important network devices running during short power failures.

SECURITY MECHANISM:

A **site-to-site VPN** should be established between the branch and headquarters to encrypt business communication. A firewall should filter unauthorized traffic. Secure wireless authentication should be used for employees.

The network can also be divided into VLANs such as:

- Employee VLAN
- Server VLAN
- Guest VLAN
- Management VLAN

This prevents guest or unauthorized devices from directly accessing sensitive company resources.

Constraint-Based Justification

Constraint	**Solution**
Rural location	4G/5G wireless connection
Limited budget	Wireless primary connection
Uninterrupted operation	Dual-WAN + satellite backup
Unauthorized access	Firewall

Secure communication	VPN

Conclusion

The proposed 4G/5G primary connection with satellite backup, dual-WAN routing, firewall, VPN, QoS and VLAN segmentation provides a reliable and secure solution for the rural branch. It balances bandwidth, cost, reliability and future scalability while supporting the company's business requirements.

2. **A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.**

ANSWER:

Problem Constraints

The major constraints are:

- More than 3,000 students.
- Large number of simultaneous users.
- Campus-wide coverage.
- Limited number of wireless access points.
- Signal interference must be minimized.
- Secure authentication is required.
- The network must provide good performance at reasonable cost.

Problem Decomposition

The design involves:

1. AP placement.
2. Frequency selection.
3. Channel planning.

4. User capacity management.
5. Authentication and security.
6. Network segmentation.

Access Point Placement

APs should be placed according to **user density rather than only coverage**.

High-density locations such as:

- Lecture halls
- Laboratories
- Libraries
- Large classrooms

should receive priority.

Hostels should have APs positioned according to building size, floor layout and expected number of users.

This prevents a single AP from becoming overloaded.

Frequency Band Selection:

- **2.4 GHz** provides longer range but has more interference and fewer channels.
- **5 GHz** should be the primary band for high-density areas because it provides greater capacity and more channels.
- Where supported, **6 GHz** can be used to provide additional capacity and reduce congestion.

Channel planning:

- Proper channel allocation is required to reduce interference.
- For 2.4 GHz, commonly used non-overlapping channels are:
- **1, 6 and 11**
- Nearby APs should be configured carefully to avoid unnecessary channel overlap.

Security and Authentication

The university should use:

- **WPA3-Enterprise**
- **802.1X authentication**
- **RADIUS server**
- This provides individual authentication for students and staff.
- Separate VLANs can be created for:

- Students
- Faculty
- Administration
- Guests
- IoT devices

Constraint-Based Justification:

CONSTRAINT	SOLUTION
3,000+ users	High-density AP planning
High traffic	5 GHz/6 GHz
Authentication	802.1X + RADIUS
2.4 GHz congestion	Band steering
Limited APs	Capacity-based placement

Conclusion

The proposed wireless network uses **strategic AP placement, 5 GHz/6 GHz prioritization, channel planning, band steering and load balancing** to support thousands of users. WPA3-Enterprise, 802.1X, RADIUS and VLANs provide secure access. The design therefore balances **coverage, performance, security and cost** while making efficient use of the limited APs.

3. **A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.**

ANSWER:
Problem Constraints
The major constraints are:

- Sensitive customer information must be protected.
- Employees require controlled access to business systems.
- Strong cybersecurity is required.
- The organization must comply with security policies.
- Network performance should not be significantly reduced.
- Operational costs should remain controlled.

Problem Decomposition

The solution can be divided into:

1. Network segmentation.
2. Firewall deployment.
3. Authentication.
4. Access control.
5. Intrusion detection and prevention.
6. Secure remote access.

Network Segmentation

The internal network should be divided into separate VLANs and security zones.

Suggested segmentation:

- Employee VLAN
- Finance VLAN
- Server VLAN
- Database VLAN
- Guest VLAN
- Management VLAN

The sensitive customer database should be isolated from normal employee and guest networks.

Firewall and DMZ

A **firewall** should be deployed at the network boundary to filter incoming and outgoing traffic.

A **DMZ (Demilitarized Zone)** should be used for public-facing services such as web servers.

A simplified architecture is:

```
Internet
   |
Edge Firewall
   |
  DMZ
   |
Internal Firewall
   |
Core Switch
   |
--------------------------------
|      |       |       |      |
Employee Finance Server Database Guest
 VLAN    VLAN    VLAN    VLAN   VLAN
```

This prevents direct Internet access to sensitive internal systems.

IDS and IPS

An **Intrusion Detection System (IDS)** monitors traffic and generates alerts about
suspicious activities.

An **Intrusion Prevention System (IPS)** can automatically block malicious

traffic.
They can detect activities such as:
- Port scanning
- Malware traffic
- Unauthorized access attempts
- Suspicious data transfers

Performance and Cost

- Security controls should be implemented without creating unnecessary network delays.
- Performance can be maintained by using appropriately sized firewalls, optimized security policies and network segmentation.

Costs can be controlled through:
- VLAN-based segmentation.
- Centralized monitoring.
- Consolidated security appliances.
- Reusing suitable existing infrastructure.

Constraint-Based Justification:

CONSTRAINT	**SOLUTION**
Sensitive data	Database VLAN
Excess privileges	RBAC
Compliance	Centralized logging
Performance	Optimized security policies
Cost	Consolidated infrastructure

Conclusion:

A segmented defense-in-depth architecture using VLANs, firewalls, DMZ, MFA,
RBAC, IDS/IPS, VPN and centralized monitoring provides strong protection for the financial institution.

The proposed design protects sensitive customer information while allowing authorized employees to access required services. It also addresses the organization's requirements for security, performance, compliance and cost-effectiveness.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, wellstructured subproblems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

		explanation			